

Acupuncture Research

Transcutaneous Electrical Acupoint Stimulation Reduces Postoperative Abdominal Pain after Endoscopic Submucosal Dissection: A Single-Center, Prospective, and Randomized Controlled Trial*

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ABSTRACT **Objective:** To evaluate the efficacy of perioperative transcutaneous electrical acupoint stimulation (TEAS) for reducing postoperative abdominal pain in patients after endoscopic submucosal dissection (ESD) for early gastric and esophageal neoplasms. **Methods:** A total of 129 patients undergoing ESD were assigned randomly to the TEAS group (64 cases) or the sham group (65 cases) using computer-generated random numbers and stratified by the surgical type (i.e., gastric or esophageal ESD). Patients in the TEAS group were treated bilaterally at the acupoints Hegu (LI 4), Neiguan (PC 6), Zusanli (ST 36), and Shangjuxu (ST 37), with the stimulation current frequency 2/100 Hz and intensity to the patient's maximum tolerance level. Patients in the sham group were treated at sham acupoints, and no electrical stimulation was given from the HANS stimulator. The TEAS and sham stimulations are both given from 30 min before ESD to the end of surgery. The primary outcome was the incidence of moderate-to-severe pain within 24 h. Secondary outcomes included the area under the curve (AUC) of Numeric Rating Scale (NRS) pain score from 1 to 48 h, morphine consumption after ESD, intraoperative hemodynamic parameters and postoperative nausea and vomiting (PONV). **Results:** The incidence of moderate-to-severe pain within 24 h postoperative period was significantly decreased by TEAS [22 (37.9%) vs. 40 (64.5%), $P=0.004$]. Patients needing morphine were also reduced by TEAS treatment [19 (32.7%) vs. 40 (64.5%), $P<0.01$]. TEAS significantly decreased the proportion and incidence of PONV compared with the sham group ($P=0.06$). Intraoperative hemodynamic parameters, recovery of gastrointestinal function, postoperative adverse events were similar between the two groups. According to the results of multivariate logistic regression, TEAS treatment was a protective factor associated with reduced incidence of NRS scores ≥ 4 (OR=0.227; 95% CI: 0.100 to 0.512; $P<0.01$), whereas patients undergoing gastric ESD had a much lower incidence of NRS scores ≥ 4 than esophageal ESD (OR=0.287; 95% CI: 0.133 to 0.621; $P=0.002$). **Conclusions:** TEAS can effectively reduce the incidence of moderate-to-severe pain in patients after ESD. TEAS demonstrates a morphine-sparing effect and reduces the occurrence of PONV. (Registration No. ChiCTR2100052837)

KEYWORDS transcutaneous electrical acupoint stimulation, endoscopic submucosal dissection, postoperative abdominal pain, randomized controlled trial

Endoscopic submucosal dissection (ESD) is a well-established minimally invasive surgery for early gastrointestinal neoplasms that has considerable advantages regarding high rate of *en bloc* removal, large and complete resection, and low risk of recurrence and fast recovery after surgery.^(1,2) Since ESD has been a widespread treatment with expanded indications, perioperative major complications such as bleeding, perforation, and stenosis have been increasingly reported.⁽²⁻⁴⁾ Postoperative abdominal pain, one of the most common ESD-related minor adverse

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events (AEs), is the main complaint from patients undergoing ESD.⁽⁵⁾ It is reported that 44.9%–62.8% of patients suffer moderate-to-severe pain after ESD, especially in the early postoperative period (within 1–4 h after surgery).^(6–8) Postoperative pain not only has a direct negative impact on patient satisfaction, but also prolongs hospitalization time and increases hospitalization costs.^(8,9) However, postoperative pain after ESD is always underestimated and ignored by anesthesiologists and endoscopists, partly due to the emphasis on enhanced recovery after surgery (ERAS) protocol on earlier discharge and earlier mobilization.⁽⁹⁾

At present, there are no unified strategies on how to alleviate postoperative pain and promote gastrointestinal function recovery for patients undergoing gastroesophageal ESD. In clinical practice, painkillers, such as opioids, may mask some postoperative complications of ESD, such as perforation, bleeding, etc.^(10,11) Some studies have found that intravenous injection of dexamethasone or lidocaine, or local infusion of bupivacaine and triamcinolone acetonide, can alleviate abdominal pain after ESD while reduce intraoperative opioid consumption and decrease patients' physical activity.^(12–14) Our previous study also found that intraoperative dexmedetomidine could significantly relieve post-ESD pain and reduce the dosage of morphine used after ESD.⁽¹⁵⁾ However, we also found that some patients still experienced abdominal distension, gastrointestinal dysfunction, or other discomfort.

Inflammatory reactions and burns caused by operational stimuli are essential in the development of postoperative pain after ESD. Electroacupuncture (EA), or transcutaneous electrical acupoint stimulation (TEAS) is a potential treatment option for postoperative pain, nausea, and vomiting.^(16,17) It can reduce inflammation via activating the vagal-adrenal pathway.⁽¹⁸⁾ Similarly, we reported that TEAS improved gastrointestinal function recovery in patients, reduced postoperative pain after abdominal surgery, and decreased the concentration of brain–gut peptide substance P.⁽¹⁶⁾ Some systematic reviews have reported the promising effects of EA or TEAS for postoperative pain management.^(19,20) However, the benefits of TEAS for patients after gastric ESD remain unclear. We therefore conducted this prospective, randomized clinical trial to assess the efficacy of TEAS for postoperative pain and gastrointestinal function recovery in patients after ESD for early gastric and

esophageal neoplasms.

METHODS

This study was a single-center, prospective, randomized, sham-controlled clinical trial conducted at Beijing Friendship Hospital affiliated to Capital Medical University in China. It was approved by the Bioethics Committee of Beijing Friendship Hospital, Capital Medical University (No. 2021-P2-315-01) and registered in the China Clinical Trials Registry (No. ChiCTR2100052837) on November 6, 2021. The study protocol for this trial was published previously.⁽²¹⁾ All participants signed the informed consent form after being recruited by the anesthesiologist at the preoperative visit.

Participants

This study was performed from December 2021 to March 2023. Inclusion criteria included: adult patients (18–75 years) with American Society of Anesthesiologists (ASA) physical status classification I to III, $18 \leq$ body mass index ≤ 30 , and scheduled to undergo elective gastric and esophageal ESD. Exclusion criteria included: patients who (1) with surgical incision or scar on the meridian of the acupoints of Hegu (LI 4), Neiguan (PC 6), Shangjuxu (ST 37), or Zusanli (ST 36); (2) local skin infection at the abovementioned acupoints; (3) nerve injury on the upper or lower limbs; (4) participated in other clinical trials within the last 4 weeks; (5) patients do not understand Numeric Rating Scale (NRS) score or unable to perform the scoring; (6) presence of a cardiac pacemaker; (7) pregnancy or in labor; (8) preoperative pain, ongoing use of central analgesic medications, and addiction or dependence on opioids; (9) severe central nervous system diseases or psychiatric disorders that may increase patient's risk or interfere with results evaluation.

Randomization and Blinding

Randomization was performed using computer-generated random numbers with a block size of 4, the allocation was sealed in an opaque envelope. Patients were randomly assigned to the sham group or TEAS group with a ratio of 1:1 by the anesthesiologist. Only the anesthesiologist and acupuncturist who was in charge of TEAS stimulation were aware of the grouping, the patients, surgeons, and physicians responsible for follow-up were unaware.

Intervention

Both TEAS and sham stimulation were performed

by a licensed acupuncturist with more than 3 years of practice. Patients in both groups received the intervention from 30 min before ESD until the end of surgery.

Patients in the TEAS group were treated bilaterally at the LI 4 and PC 6, ST 36 and ST 37. The detail of the location of acupoints are presented in the Appendix 1. Self-adhesive electrodes with wires were attached to the location of these 4 acupoints and connected to the HANS acupoint nerve stimulator (HANS-200A, Nanjing Jisheng Medical Technology Co., Ltd., China), with the current frequency of 2/100 Hz (wave width: 2 Hz, 0.6 ms; 100 Hz, 0.2 ms), alternating frequency for 3 s, and stimulation intensity was set to the patient's maximum tolerance level. In the sham group, 2 of the sham points were located at 7 cun above and 1 cun outside Shenmen (HT 7). The other 2 sham points were at 9 cun and 12 cun above Kunlun (BL 60, Appendix 2). Self-adhesive electrodes were glued to these points, but no electrical stimulation was given from the HANS stimulator.

Anesthesia Procedures

On the day before ESD, all enrolled patients signed informed consent and received relevant introduction on the implementation and follow-up. Electrocardiogram, blood pressure, pulse oxygen saturation, and bispectral (BIS) were monitored before anesthesia. TEAS and sham stimulation were conducted by the experienced acupuncturist 30 min before ESD. Midazolam (5 mg, 0.03 mg/kg; Jiangsu Ehwa Pharmaceutical Co., Ltd., TMD25B01), remifentanyl (1 mg, 1–2 μ g/kg; Yichang Renfu Pharmaceutical Co., Ltd., AD5030381), etomidate (20 mg, 0.1–0.2 mg/kg; Jiangsu Ehwa Pharmaceutical Co., Ltd., TYT24K18) and rocuronium (25 mg, 0.6–0.8 mg/kg; Hainan STAR Pharmaceutical Co., Ltd., 2412045A) were used before intubation. Anesthesia was maintained with a total intravenous infusion of propofol [200 mg, 4–6 mg/(kg·h); Beijing Fresenius Kabi Pharmaceutical Co., Ltd., 16TK1957], remifentanyl [0.05–0.2 μ g/(kg·min)], and rocuronium (one-third of the induced dose every 40 min). Anesthetic depth was controlled by the adjustment of intravenous propofol to maintain a BIS index of 45–55 during the procedure. A total of 50 mg tramadol (100 mg, Grunenthal GmbH, 00406X) was applied for analgesia 30 min before the end of ESD. At the end of the surgery, the patients were extubated and transferred to the post-anesthesia care unit in routine.

NRS was used to assess the patient's pain level. If the NRS score ≥ 4 points or the patient requested analgesia during the follow-up, 1 mg morphine (Northeast Pharmaceutical Group, Shenyang First Pharmaceutical Co., Ltd., 241103) was administered intravenously. Intraoperative mean arterial pressure (MAP), heart rate (HR), SpO₂, dose of drugs as well as fluid input and output were recorded. The concomitant medication during the trial period from the start of surgery to 2 days after the end was recorded.

Primary and Secondary Outcomes

A trained research assessor who blinded to the group allocation performed the assessment of primary and secondary outcomes. The primary outcome was the incidence of moderate-to-severe pain evaluated by NRS score within 24 h postoperative period. The scale was 0–10: 1–3 for mild pain, 4–6 for moderate pain, and 7–10 for severe pain. The moderate-to-severe post-ESD pain was defined as NRS ≥ 4 .^(15,16)

The secondary outcomes included the NRS pain scores at 1, 2, 4, 6, 18, 24, and 48 h postoperatively; the consumption of morphine, gastrointestinal function recovery time, the incidence of postoperative complications, and patient satisfaction during follow-up period. A scale with a maximum score of 100 was utilized to evaluate patient satisfaction after ESD. Zero point indicated extreme dissatisfaction, and 100 points indicate very satisfaction, with patients scoring verbally.

For recovery of gastrointestinal function, we mainly recorded the time of first flatus, defecation, drink, and solid food intake of patients. The main recorded postoperative complications were hypertension, hypotension, tachycardia, bradycardia, postoperative nausea and vomiting (PONV), gastrointestinal perforation or bleeding, reoperation, and fever. Postoperative AEs were recorded by the follow-up anesthesiologist, and in the event of a malignant event such as bleeding >500 mL, the trial will be immediately terminated and reported to the principal investigator and a review board will be established.

Sample Size Estimation

PASS 2011 software (NCSS, USA) was used to estimate the sample size. The sample size was calculated based on the outcomes of our pre-trial. In the pre-trial, we observed that the incidence of moderate-to-severe pain after ESD was 64%. We hypothesized that TEAS could reduce the incidence

of moderate-to-severe pain by about 40%. Setting an α of 0.05, a β of 0.1, and a power of 0.8, plus a 10% dropout rate, we calculated that a total of 120 cases were needed, 60 cases per group were required.

Statistical Analysis

All data were analyzed using SPSS 26.0 (IBM Corp., USA). Continuous variables were reported as mean $\pm$ standard deviation ($\bar{x} \pm s$) or median (interquartile range, IQR). For normally distributed data, differences between groups were analyzed by *t*-test, and skewed data by Mann-Whitney *U* test. Categorical variables were reported as the number of cases (percentages) and intergroup differences were analyzed by Chi-square test or Fisher's test. A *P*-value less than 0.05 indicates a statistically significant difference.

Predefined subgroup analyses were performed to assess differences in the primary outcome according to gender, type of surgery, and tumor size. The absolute standard mean difference (ASD) was calculated between groups to define the absolute difference in means, mean ranks or proportions. Any variables with $ASD > 0.36$ ($1.96 \times \sqrt{\frac{1}{60} + \frac{1}{60}} = 0.36$) were defined as unbalanced and were adjusted based on primary outcome. To calculate ordinal variables, such as NRS pain score, the distribution bar was used to demonstrate the severity of pain.

Perioperative factors were analyzed using univariate and multivariate logistic regression analysis, and the resulting risk ratio (RR) values, 95% confidence interval (CI), and *P* values were provided. The potential influencing factors included in this trial were derived from previous investigations.^(7,8) All perioperative factors that may cause NRS scores ≥ 4 were analyzed with univariate logistic regression analysis.

RESULTS

Patients and Surgery Characteristics

From December 2021 to March 2023, a total of 275 patients were assessed for eligibility. Among them, 11 patients did not meet the inclusion criteria, 3 had ASA Physical Status Classification grades $> III$, 125 had took part in earlier research, 2 with cardiac pacemakers, and 5 had opted out of the trial. As a result, 129 patients were enrolled and randomized assigned to the TEAS or the sham group. One patient from the TEAS group and 1 from the sham group were converted

to laparotomy due to further pathological infiltration of lesions and were eliminated from the study. Two patients from the TEAS group and 1 from the sham group dropped out due to postoperative bleeding. Three patients from the TEAS group and 1 from the sham group withdrew from the trial. Finally, a total of 120 patients completed the entire research, 58 in the TEAS group and 62 in the sham group (Figure 1).

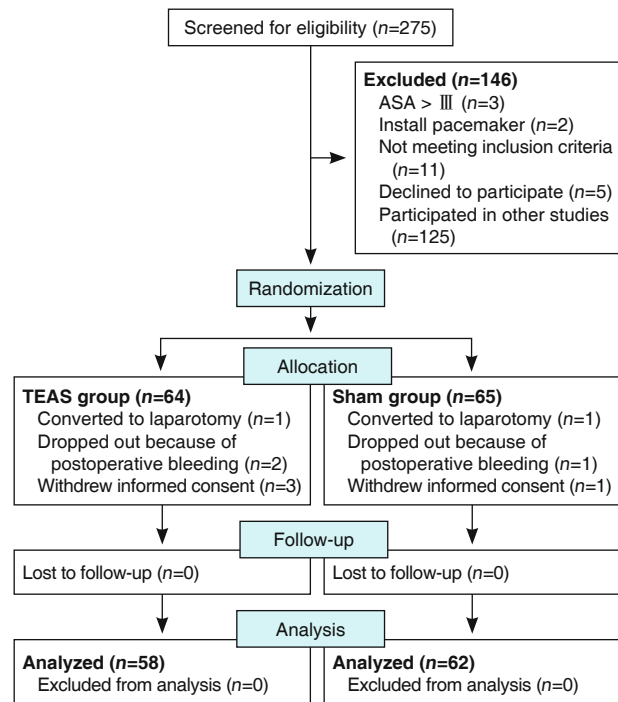


Figure 1. Recruitment Flow Chart for Patients Undergoing Endoscopic Submucosal Dissection

Notes: TEAS: transcutaneous electrical acupoint stimulation; ASA: American Society of Anesthesiologists. The same below

The baseline characteristics of 120 patients are detailed in Table 1. There were no statistically significant differences in the baseline characteristics of the patients across groups. No difference in surgery duration, anesthesia duration, consumption of anesthetics and infusion volume were found ($P > 0.05$).

Primary Outcome

The incidence of moderate-to-severe pain in the TEAS group was 22/58 (37.9%), whereas in the sham group was 40/62 (64.5%, RR 0.588; 95% CI, 0.403–0.858; $P=0.004$, Table 2). This difference was more significant in esophageal ESD, the incidence was 13/29 (44.8%) in the TEAS group and 25/29 (86.2%) in the sham group.

Secondary Outcomes

The NRS pain score was lowered by TEAS for the

first 24 h after ESD ($P=0.001$), whereas the difference was not significant at 48 h (Table 2). As shown in Figure 2, we also found that only 12.1% patients in TEAS group experienced moderate pain (NRS 4–6), whereas 48.4% in the sham group ($P<0.01$). The acute pain was improved at 2 to 6 h after ESD in both groups. But at 18 h, there was 35.5% patients complained moderate pain in sham group, whereas 3.4% in the TEAS group ($P<0.01$). No difference was found at 48 h post-ESD between 2 groups ($P>0.05$). The patients needed morphine to relieve post-ESD pain was also decreased by TEAS treatment compared with the sham group (32.7% vs. 64.5%, $P<0.01$, Table 2). No patient experienced severe pain in both groups.

There was no statistically significant difference between the two groups in the postoperative recovery of gastrointestinal function ($P>0.05$, Table 2). The incidence of PONV were significantly lower than in the sham group (6.9% vs. 22.6%, $P=0.016$). No differences in other postoperative AEs were found between the 2 groups ($P>0.05$).

Associations between Perioperative Influence Factors and Pain Score

As shown in Table 2, there were 62 (51.7%) cases experienced moderate-to-severe pain totally. The results of univariate logistic regression analysis showed that TEAS treatment ($P=0.004$), type of ESD surgery ($P=0.003$), and tumor location ($P=0.032$) were potential contributing factors associated with NRS scores ≥ 4 (Appendix 3). Therefore, we performed multivariate logistic regression analysis for all factors with $P<0.1$ (Figure 3). Forest plot showed that TEAS treatment was a protective factor associated with reduced incidence of NRS scores ≥ 4 (OR=0.227; 95% CI: 0.100 to 0.512; $P<0.01$). In addition, the type of ESD surgery was also a protective factor. Patients undergoing gastric ESD had a much lower incidence of NRS scores ≥ 4 than patients undergoing esophageal ESD (OR=0.287; 95% CI: 0.133 to 0.621; $P=0.002$). The subgroup analysis indicated that the primary outcome was not affected by gender, type of surgery, or tumor area ($P>0.05$, Appendix 4).

Patient Satisfaction

In TEAS and sham groups, the mean ranks of patient satisfaction scores were 73.0 ± 28.5 and 48.8 ± 21.2 , respectively. In comparison with the sham group, patient satisfaction in the TEAS group was considerably higher ($P<0.01$).

Table 1. Characteristics of Patients Undergoing ESD at Baseline and Intra-operative Period [Case (%)]

Characteristics	TEAS group (58 cases)	Sham group (62 cases)	ASD
Age [Year, Median (IQR)]	65 (54, 69)	62 (53, 67)	0.144
BMI [kg/m ² , Median (IQR)]	24.0 (21.5, 26.0)	24.0 (22.5, 25.9)	0.032
Female	20 (34.5)	18 (29.0)	0.117
Type of surgery			
Gastric ESD	29 (50.0)	33 (53.2)	0.065
Esophageal ESD	29 (50.0)	29 (46.8)	0.065
Lesion location			
Gastric cardia	10 (17.2)	9 (14.5)	0.019
Gastric body	6 (10.3)	6 (9.7)	0.067
Gastric antrum	10 (17.2)	13 (21.0)	0.077
Gastric angle	3 (5.2)	5 (8.1)	0.081
Upper esophagus	2 (3.4)	3 (4.8)	0.062
Median esophagus	20 (34.5)	20 (32.3)	0.037
Lower esophagus	7 (12.1)	6 (9.7)	0.085
Histopathology			
Benign tumor	7 (12.1)	8 (12.9)	0.158
Dysplasia	11 (19.0)	14 (22.6)	0.000
Adenoma	25 (43.0)	24 (38.7)	0.000
Carcinoma	15 (25.9)	16 (25.8)	0.097
Tumor area $\geq 1,000$ mm ²	22 (37.9)	27 (43.5)	0.133
Multiple lesions	3 (5.2)	10 (16.1)	0.347
Smoking history	27 (46.5)	32 (51.6)	0.085
History of alcohol intake	31 (53.4)	33 (53.2)	0.019
Previous surgical history	38 (65.5)	50 (80.7)	0.215
ASA physical status classification			
I	9 (15.5)	14 (22.6)	0.069
II	40 (69.0)	37 (59.7)	0.105
III	9 (15.5)	11 (17.7)	0.054
Comorbidities			
Hypertension	29 (50.0)	33 (53.2)	0.077
Diabetes	11 (19.0)	11 (17.7)	0.035
Coronary disease	5 (8.6)	8 (12.9)	0.210
Hyperlipidemia	22 (37.9)	20 (29.4)	0.037
Cerebral infarction	3 (5.2)	11 (17.7)	0.157
Dose of anesthetic inducers [$\bar{x} \pm s$ /Median (IQR)]			
Midazolam (mg)	2 (1.8, 2.2)	2 (1.8, 2.2)	0.074
Remifentanyl (μ g, $\bar{x} \pm s$)	69 $\pm$ 11	70 $\pm$ 9	0.007
Rocuronium bromide (mg)	50 (42, 50)	50 (42, 50)	0.244
Etomidate (mg)	14 (12, 18)	15 (13, 20)	0.339
Dose of anesthetic maintenance drugs [Median (IQR)]			
Propofol (mg)	340 (255, 480)	420 (235, 525)	0.247
Remifentanyl (μ g)	672 (519, 884)	805 (435, 1088)	0.178
Duration of surgery [min, Median (IQR)]	76 (53, 112)	76 (43, 112)	0.129
Duration of anesthesia [min, Median (IQR)]	85 (63, 120)	85 (55, 120)	0.134
Intraoperative infusion volume [mL, Median (IQR)]	500 (400, 650)	500 (400, 650)	0.005

Notes: The absolute standard mean difference (ASD) is calculated between groups to define the absolute difference in means, mean ranks, or proportions divided by the pooled SD. ASD more than 0.36 were defined as imbalanced. IQR: interquartile range; ESD: endoscopic submucosal dissection

Table 2. Primary and Secondary Outcomes of Patients Undergoing ESD after Treatment

Variables	TEAS group (58 cases)	Sham group (62 cases)	RR/mean difference (95% CI)	P value
Primary outcome [Case (%)]				
Incidence of moderate-to-severe pain	22 (37.9)	40 (64.5)	0.588 (0.403–0.858)	0.004 [‡]
Secondary outcomes				
Postoperative NRS pain score ($\bar{x} \pm s$ /Median (IQR))				
1 h	1.3 $\pm$ 0.5	2.6 $\pm$ 2.9	−1.4 (−2.3 to 1.2)	0.000 [*]
2 h	1.6 (0.6, 2.0)	2.1 (0.5, 3.2)	−0.5 (−0.9 to 0.4)	0.013 [†]
4 h	1.6 $\pm$ 1.3	2.0 $\pm$ 1.5	−0.4 (−0.8 to 1.2)	0.165 [*]
6 h	1.5 (0.6, 2.7)	2.7 (1.1, 4.3)	−1.2 (−1.8 to 0.4)	0.023 [†]
18 h	1.5 $\pm$ 1.2	3.2 $\pm$ 1.5	−1.7 (−2.0 to 0.4)	0.000 [*]
24 h	1.2 $\pm$ 1.3	2.8 $\pm$ 1.9	−1.6 (−3.0 to 1.8)	0.001 [*]
48 h	0.5 $\pm$ 0.4	0.9 $\pm$ 0.6	−0.4 (−1.1 to 0.2)	0.635 [*]
Patients needing morphine [Case (%)]	19 (32.7)	40 (64.5)	0.508 (0.336 to 0.767)	0.000 [‡]
Nausea and vomiting [Case (%)]	4 (6.9)	14 (22.6)	0.840 (0.415 to 1.698)	0.016 [‡]
Recovery of gastrointestinal function [h, $\bar{x} \pm s$ /Median (IQR)]				
Time to first flatus	11.2 (9.5, 12.8)	14.5 (6.2, 23.0)	−3.3(−6.5 to 1.2)	0.738 [†]
Time to first defecation	48.0 $\pm$ 23.6	54.2 $\pm$ 30.1	−6.2 (−8.1 to 2.3)	0.318 [*]
Time to first drinking	69.2 $\pm$ 22.1	68.5 $\pm$ 11.6	1.2 (−1.3 to 2.5)	0.766 [*]
Time to first eating	73.1 $\pm$ 12.1	75.3 $\pm$ 13.3	−2.6 (−4.5 to 2.1)	0.350 [*]
Patients' satisfaction (score, $\bar{x} \pm s$)	73.0 $\pm$ 28.5	48.8 $\pm$ 21.2	−24.5 (−30.1, 6.4)	0.000 [*]

Notes: RR: risk ratio. NRS: Numeric Rating Scale. The same below. ^{*}P value is obtained by Student's *t*-test. [†]P value is obtained by Mann-Whitney *U* test. [‡]P value is obtained by χ^2 analysis

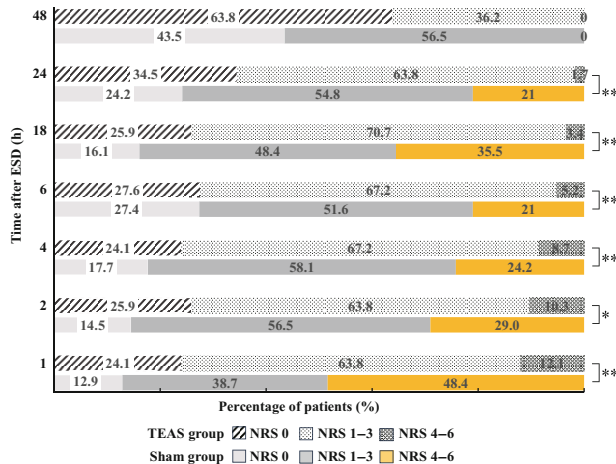


Figure 2. Numeric Rating Scale Pain Score Distribution in Patients Undergoing ESD between Two Groups

Notes: ^{*}P<0.05, ^{**}P<0.01

DISCUSSION

In this study, our data suggested that in patients undergoing gastroesophageal ESD under general anesthesia, pretreating with TEAS could effectively improve the early postoperative pain of patients, reduce the incidence of moderate-to-severe pain, and decrease the application of morphine after surgery. In addition, the results showed that the type of surgery was a risk factor

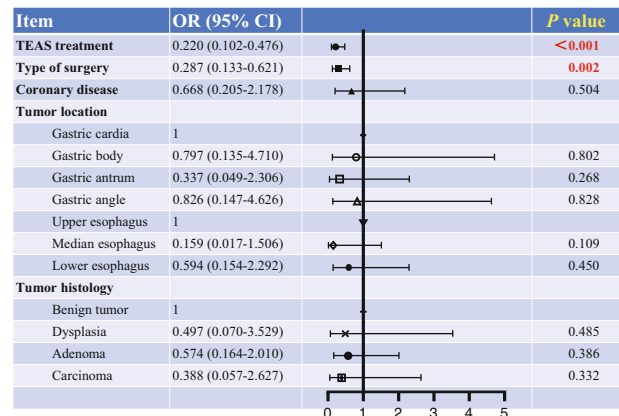


Figure 3. Forest Plot of Multivariate Logistic Regression Analysis of Risk Factors for Moderate-to-Severe Pain in Patients Undergoing ESD

for postoperative pain. Patients undergoing gastric ESD had a much lower incidence of NRS scores ≥ 4 than patients undergoing esophageal ESD. Moreover, this treatment could reduce the occurrence of PONV, and correspondingly achieve higher patient satisfaction.

Based on recently published data, the incidence of moderate-to-severe pain induced by ESD was as high as 44.9% to 62.8%.⁽⁶⁻⁸⁾ The same conclusion was obtained in Jung's study.⁽²²⁾ In our research, we found that the incidence of moderate-to-severe pain in the

sham group was 64.5%. It was further demonstrated that the rate of post-ESD pain is high enough for physicians to pay more attention and make active pain management. However, there are currently no unified measure on how to carry out good postoperative analgesia for patients undergoing gastroesophageal ESD. Our study found that TEAS can effectively alleviate post-ESD pain. In this regard, several recently published investigations confirmed that TEAS can provide powerful protection against postoperative pain for a variety of procedures, like gastric and colorectal surgery⁽¹⁶⁾ as well as pediatric orthopedic surgery.⁽²³⁾ More importantly, we also found that pretreating with TEAS was a protective factor associated with reduced incidence of moderate-to-severe pain according to logistic regression analysis.

Based on published data, post-ESD pain is more severe in the early postoperative period (within 1–4 h after surgery).⁽⁶⁻⁸⁾ In fact, this was also confirmed in our preliminary experiment.⁽²¹⁾ In line with these results, we followed up the NRS scores of patients in the early postoperative period. Moreover, we appropriately extended the visit to 48 h after surgery. Further investigation showed that TEAS could improve post-ESD pain within 24 h after surgery. From the results of the distribution graph of NRS, at 1 h after surgery, we found that TEAS reduced the percentage of patients with moderate pain to 12.1%, while the sham group was 48.4%. At 2, 4 and 6 h after ESD, the patients with moderate pain in both groups were reduced, but the proportion of patients with moderate pain in the TEAS group was lower than that in the sham group. This result was exactly corresponding to the patients who were given morphine treatment, because at the beginning of the trial design, patients with moderate pain would be given morphine treatment. Except for some patients who refused to use morphine, most patients received morphine treatment when NRS >4 for the first time, and then the pain was relieved. TEAS decreased the percentage of NRS >4 at 18 and 24 h. At 48 h post-ESD, the difference was not significant. This result revealed that the adjuvant analgesic effect of perioperative TEAS could last until 24 h after ESD.

In addition, a recently published retrospective study confirmed that gender, operation duration, and lesion site were influence factors affecting postoperative pain in ESD.⁽⁸⁾ Strikingly, our study revealed that patients undergoing gastric ESD had a much lower incidence of moderate-to-severe pain than esophageal ESD.

This may be due to the different mechanisms of pain transmission in different organs. Moreover, we cannot rule out the possibility that increased peristalsis at the esophageal site and frequent contractions may also lead to aggravated postoperative pain after ESD.

In this study, peri-operative TEAS treatment decreased the cases of PONV, this result is similar to our previous study in gastrointestinal surgery.⁽¹⁶⁾ As known, LI 4, PC 6, ST 36, and ST 37 are the most commonly used acupoints for regulating the intestines and stomach, channeling the channels and collaterals. It was well documented that stimulating these acupoints could improve intestinal motility, and regulate endocrine function, which helps reduce PONV.⁽²⁴⁾ However, the data from this study failed to confirm that TEAS improved gastrointestinal function in patients after gastroesophageal ESD. This result is inconsistent with another study by our team.⁽¹⁶⁾ In this regard, we consider that ESD is a minimally invasive surgery in nature, it has little interference with gastrointestinal function, so it has almost no impact on the exhaust function of patients. What is more, the eating and drinking time of patients after ESD surgery is basically carried out according to the doctor's advice, the time of all patients is therefore relatively fixed.

There are several limitations in this study. Firstly, TEAS was performed only during the operation, and there was no postoperative management. Although, in our previous observation, we found that the postoperative pain was most severe in the early postoperative period for most patients. Hence, TEAS was not immediate administered after patients returned to the ward. However, we still cannot rule out that this may be related to the lack of difference in pain 1 h after ESD between the TEAS and the sham groups. Secondly, some analgesic substances such as endorphins, or some injury-related factors such as interleukin-6 to quantitatively compare the postoperative pain response between the two groups were not included in the study due to limited research fund.

In conclusion, data from this study showed that pretreating with TEAS could effectively improve the early postoperative pain of patients undergoing ESD, and reduce the incidence of moderate-to-severe pain, providing a feasible strategy for relieving the condition.

Conflict of Interest

None declared.

Author Contributions

An LX is the chief investigator who is responsible for quality control and study design, trial overseeing, and paper revision. Liu XM and Qi YX recruited participants, analyzed the data, improved the study protocol, and drafted this manuscript. Zheng JY, Hao SQ, Hao WW, Du Y and Li WJ worked for data collection and performed anesthesia and intervention. All authors have read and approved the final version for publication.

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